

Chapter: Introduction

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Exercise 1.12

Pattern Recognition and Machine Learning - Christopher M. Bishop

Solution

In this exercise, we are asked to show that

$$\mathbb{E}[x_n x_m] = \mu^2 + I_{nm} \sigma^2$$

where x_n and x_m denote data points sampled from a Gaussian distribution with mean μ and variance σ^2 , and I_{nm} satisfies $I_{nm} = 1$ if $n = m$ and $I_{nm} = 0$ otherwise.

To prove this, we start by considering the definition of the expectation of the product of two random variables:

$$\mathbb{E}[x_n x_m] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_n x_m p(x_n, x_m) dx_n dx_m$$

We know that if $n \neq m$, then x_n and x_m are independent, and thus the joint probability density function can be expressed as the product of their individual densities:

$$p(x_n, x_m) = p(x_n)p(x_m)$$

Then, we can write:

$$\begin{aligned} \mathbb{E}[x_n x_m] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_n x_m p(x_n) p(x_m) dx_n dx_m \\ &= \int_{-\infty}^{\infty} x_n \left(\int_{-\infty}^{\infty} x_m p(x_m) dx_m \right) p(x_n) dx_n \\ &= \int_{-\infty}^{\infty} x_n \mathbb{E}[x_m] p(x_n) dx_n \\ &= \mathbb{E}[x_m] \int_{-\infty}^{\infty} x_n p(x_n) dx_n \\ &= \mathbb{E}[x_m] \mathbb{E}[x_n] \\ &= \mathbb{E}[x_n] \mathbb{E}[x_m] \\ &= \mu^2 \end{aligned}$$

Now, if $n = m$, then we have:

$$\begin{aligned} \mathbb{E}[x_n x_n] &= \mathbb{E}[x_n^2] \\ &= \text{Var}(x_n) + (\mathbb{E}[x_n])^2 \\ &= \sigma^2 + \mu^2 \end{aligned}$$

We can conclude that:

$$\mathbb{E}[x_n x_m] = \mu^2 + I_{nm} \sigma^2$$

where $I_{nm} = 1$ if $n = m$ and $I_{nm} = 0$ otherwise, which completes the proof.

Expected maximum-likelihood mean: result (1.57). The estimators are

$$\mu_{ML} = \frac{1}{N} \sum_{n=1}^N x_n, \quad \sigma_{ML}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{ML})^2.$$

Both are random variables because they depend on the sample. The true parameters μ and σ^2 are fixed, and expectations are taken over the possible samples. For the mean,

$$\begin{aligned} \mathbb{E}[\mu_{ML}] &= \mathbb{E} \left[\frac{1}{N} \sum_{n=1}^N x_n \right] \quad (\text{definition of } \mu_{ML}) \\ &= \frac{1}{N} \sum_{n=1}^N \mathbb{E}[x_n] \quad (\text{linearity of expectation}) \\ &= \frac{1}{N} \sum_{n=1}^N \mu \quad (\text{each } x_n \text{ has mean } \mu) \\ &= \mu \quad (\text{there are } N \text{ identical terms}) \end{aligned}$$

Expected maximum-likelihood variance: result (1.58). Expanding the squared residual gives

$$\begin{aligned} \mathbb{E}[\sigma_{ML}^2] &= \mathbb{E} \left[\frac{1}{N} \sum_{n=1}^N (x_n - \mu_{ML})^2 \right] \\ &= \frac{1}{N} \sum_{n=1}^N \mathbb{E}[(x_n - \mu_{ML})^2] \\ &= \frac{1}{N} \sum_{n=1}^N (\mathbb{E}[x_n^2] - 2\mathbb{E}[x_n \mu_{ML}] + \mathbb{E}[\mu_{ML}^2]) . \end{aligned}$$

We already know that $\mathbb{E}[x_n^2] = \mu^2 + \sigma^2$. We now calculate the other two expectations explicitly.

1. Calculate $\mathbb{E}[x_n \mu_{ML}]$. Fix n . Since μ_{ML} includes x_n , they are not independent. Substituting the sample mean and separating the term $m = n$ gives

$$\begin{aligned} \mathbb{E}[x_n \mu_{ML}] &= \mathbb{E} \left[x_n \frac{1}{N} \sum_{m=1}^N x_m \right] \\ &= \frac{1}{N} \sum_{m=1}^N \mathbb{E}[x_n x_m] \\ &= \frac{1}{N} \left(\mathbb{E}[x_n^2] + \sum_{\substack{m=1 \\ m \neq n}}^N \mathbb{E}[x_n x_m] \right) \\ &= \frac{1}{N} ((\mu^2 + \sigma^2) + (N-1)\mu^2) \\ &= \frac{1}{N} (N\mu^2 + \sigma^2) \\ &= \mu^2 + \frac{\sigma^2}{N}. \end{aligned}$$

There is one diagonal term and $N - 1$ terms involving distinct samples. In particular, replacing $\mathbb{E}[x_n \mu_{ML}]$ by $\mathbb{E}[x_n] \mathbb{E}[\mu_{ML}] = \mu^2$ would be incorrect.

2. Calculate $\mathbb{E}[\mu_{ML}^2]$. The square of a sum contains all ordered pairs of indices:

$$\begin{aligned}\mu_{ML}^2 &= \left(\frac{1}{N} \sum_{n=1}^N x_n \right) \left(\frac{1}{N} \sum_{m=1}^N x_m \right) \\ &= \frac{1}{N^2} \sum_{n=1}^N \sum_{m=1}^N x_n x_m.\end{aligned}$$

Of the N^2 ordered pairs, N have $n = m$ and $N(N - 1)$ have $n \neq m$. Therefore,

$$\begin{aligned}\mathbb{E}[\mu_{ML}^2] &= \frac{1}{N^2} \sum_{n=1}^N \sum_{m=1}^N \mathbb{E}[x_n x_m] \\ &= \frac{1}{N^2} \left(\sum_{n=1}^N \mathbb{E}[x_n^2] + \sum_{n=1}^N \sum_{\substack{m=1 \\ m \neq n}}^N \mathbb{E}[x_n x_m] \right) \\ &= \frac{1}{N^2} (N(\mu^2 + \sigma^2) + N(N - 1)\mu^2) \\ &= \frac{1}{N^2} (N\mu^2 + N\sigma^2 + (N^2 - N)\mu^2) \\ &= \frac{1}{N^2} (N^2\mu^2 + N\sigma^2) \\ &= \mu^2 + \frac{\sigma^2}{N}.\end{aligned}$$

Notice that $\mathbb{E}[\mu_{ML}^2] \neq (\mathbb{E}[\mu_{ML}])^2$; their difference is $\text{Var}(\mu_{ML}) = \sigma^2/N$.

3. Substitute and simplify. For each n , the expected squared residual is

$$\begin{aligned}\mathbb{E}[(x_n - \mu_{ML})^2] &= \mathbb{E}[x_n^2] - 2\mathbb{E}[x_n \mu_{ML}] + \mathbb{E}[\mu_{ML}^2] \\ &= (\mu^2 + \sigma^2) - 2\left(\mu^2 + \frac{\sigma^2}{N}\right) + \left(\mu^2 + \frac{\sigma^2}{N}\right) \\ &= \mu^2 + \sigma^2 - 2\mu^2 - \frac{2\sigma^2}{N} + \mu^2 + \frac{\sigma^2}{N} \\ &= (\mu^2 - 2\mu^2 + \mu^2) + \sigma^2 - \frac{2\sigma^2}{N} + \frac{\sigma^2}{N} \\ &= \sigma^2 - \frac{\sigma^2}{N} \\ &= \frac{N - 1}{N} \sigma^2.\end{aligned}$$

Finally, averaging these N identical expectations gives

$$\begin{aligned}\mathbb{E}[\sigma_{ML}^2] &= \frac{1}{N} \sum_{n=1}^N \mathbb{E}[(x_n - \mu_{ML})^2] \\ &= \frac{1}{N} \sum_{n=1}^N \frac{N-1}{N} \sigma^2 \\ &= \frac{1}{N} \cdot N \cdot \frac{N-1}{N} \sigma^2 \\ &= \boxed{\frac{N-1}{N} \sigma^2}.\end{aligned}$$

Q.E.D.